MID-TERM EXAMINATION

B. Tech (AI&ML/CSE-AI/CSE/IT) - Semester: 4th

(March 2026) OFFLINE mode

Subject Code: BIT 204	Subject: Operating Systems
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3.

		(2.5*4=10)	CO Mapping
Q1			
a)	Contrast user-level threads with kernel-level threads. Why is a blocking system call problematic for user-level threads?	2.5	CO1
b)	Explain Process Life Cycle in detail with diagram.	2.5	CO1
c)	Describe the significance of the valid/invalid bit in paging systems. How does it assist in handling page faults?	2.5	CO2
d)	Define critical section. State three conditions of the Critical Section problem.	2.5	CO2
	UNIT I		**CO Mapping**
Q2	**Attempt any two parts**	**(5*2=10)**	
a)	Consider the following set of processes: **Process** / **Arrival Time** / **Burst Time** P1 / 0 / 5 P2 / 1 / 4 P3 / 2 / 2 P4 / 4 / 6 P5 / 6 / 3 The CPU scheduling policy is **Round Robin** with **Time Quantum = 3 units**. Answer the following: I. Draw the **Gantt chart.** II. Calculate the **Average Waiting Time (AWT).** III. Calculate the **Average Turnaround Time (ATT).**	5	CO1
b)	Explain the following with diagram: i. Single Threaded vs Multithreaded Programming ii. Process Control Block (PCB)	5	CO1
c)	Explain the distinct roles of a short-term scheduler, medium term scheduler and a long-term scheduler with diagram.	5	CO1
	UNIT II		**CO Mapping**
Q3	**Attempt any two parts**	**(5*2=10)**	
a)	Explain the concept of Segmentation in memory management. Compare it briefly with paging to highlight why segmentation is considered more user-oriented.	5	CO2
b)	A system has 4 processes (P0–P3) and 3 resource types (A, B, C). The total instances of resources are: $A = 8, B = 6, C = 5$	5	CO2

	The current system snapshot is given below: Process — Allocation (A B C) — Max (A B C) P0 — 1 2 0 — 3 3 2 P1 — 2 0 1 — 4 2 2 P2 — 1 1 1 — 2 2 3 P3 — 1 0 2 — 3 1 3 Answer the following using Banker's Algorithm: I. Compute the Available vector and the Need matrix. II. Is the system in a safe state? If yes, write one safe sequence.			
c)	A system has the following memory blocks (in KB): 100, 500, 200, 300, 600 The following processes (in KB) arrive in order: $P1 = 212, P2 = 417, P3 = 112, P4 = 426$ Allocate memory to the processes using the following allocation strategies First Fit, Best Fit, Worst Fit and Next Fit. For each method Show the allocation of processes to memory blocks, Indicate which processes (if any) cannot be allocated and show the remaining memory in each block after allocation.	5		